
Time: Three Hours

Maximum Marks: 75

SECTION - A ($10 \times 2 = 20$ Marks)**Answer Any TEN Questions**

1. State the axioms of probability.
2. Define conditional probability.
3. Define a discrete random variable. Give an example for the same.
4. Define the probability density function of the continuous random variable X .
5. Recall the definition of the expected value of X , when X is a discrete random variable.
6. Write down the mean and variance of binomial distribution with parameters n and p .
7. State the area property of the normal distribution.
8. List any two situations where Poisson probability law can be applied.
9. State the central limit theorem.
10. Describe about confidence interval.
11. Write any two properties of the sampling distribution of t -test.
12. Mention any two uses of χ^2 test.

SECTION - B ($5 \times 5 = 25$ Marks)**Answer Any FIVE Questions**

13. If A and B are two events in a sample space S , then prove that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$
14. Two boxes contain 12 white and 18 black and 15 white and 25 black balls respectively. One box was taken at random and a ball was taken from the same. It is a black ball. What is the probability that it is from the first box?
15. If X and Y are independent random variables defined on a sample space S , then prove that $E(XY) = E(X)E(Y)$.
16. Determine the binomial distribution for which the mean is 4 and variance 3. Also find $P(X = 15)$.
17. Write the step-by-step procedure for Test of Hypothesis.
18. A sample of ten house owners is drawn and the following values of their incomes are obtained. Mean Rs.6,000; Standard deviation Rs.650. Test the hypothesis that the average income of house owners of the town is Rs.5500.
19. From the following data test if the difference between the variances is significant at 5% level of significance.

<i>Sum of the squares of squares of deviations from the mean</i>	84.4	102.6
<i>Size</i>	8	10
<i>Sample</i>	A	B

SECTION - C ($3 \times 10 = 30$ Marks)**Answer Any THREE Questions**

20. There are three boxes containing respectively 1 white, 2 red, 3 black balls; 2 white, 3 red, 1 black ball; 3 white, 1 red and 2 black balls. A box is chosen at random and from it two balls are drawn at random. The two balls are 1 red and 1 white. What is the probability that they come from (i) the first box (ii) second box and (iii) third box.
21. A random variable X has the following probability function.

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find the value of k and evaluate the following:

- (i) $P(X < 6)$; (ii) $P(X \geq 6)$; (iii) $P(0 < X < 5)$.
22. Life of a certain kind of electronic device has a mean of 300 hours and a standard deviation of 25 hours. Assume that the distribution of life times which are measured to the nearest hour can be approximated closely with a normal curve. (i) Find the probability that any one of these devices will have a life time of more than 350 hours. (ii) What percent will have life time from 220 to 260 hours?
23. The average number of defective articles per day in a certain factory is claimed to be less than the average of all the factories. The average of all the factories is 30.5. A random sample of 100 days showed the following distribution.

Class limits	6-20	21-25	26-30	31-35	36-40
No. of days	12	22	20	30	16

Is the average less than the figure for all the factories?

24. What is meant by Analysis of Variance? Explain in detail about one-way classification and two-way classification of analysis of variance.